

Email Header Forensic Analysis using Python

Python Program:

```
raw_header = '''Delivered-To: victim@example.com
Received: from mail-relay-99.suspicious-host.ru (unknown [45.33.32.156])
by mx.example.com
From: "Bank Support" <support@paypal-secure-verify.com>
Reply-To: attacker@totallynotscam.net
Subject: Urgent: Verify your account'''

def analyze_header(header_text):
    findings=[]
    for line in header_text.splitlines():
        if line.startswith("Received:") and "suspicious" in line.lower():
            findings.append("Suspicious relay server found")
        if line.startswith("Reply-To:"):
            findings.append("Reply-To differs from sender")
        if line.startswith("From:") and "-secure-verify" in line:
            findings.append("Sender domain uses suspicious naming")
    return findings

print("Email Header Forensic Analysis")
for finding in analyze_header(raw_header):
    print("-", finding)
```

Sample Output:

```
Email Header Forensic Analysis
- Suspicious relay server found
- Sender domain uses suspicious naming
- Reply-To differs from sender - possible spoofing
```

Explanation:

This program demonstrates a simple email header forensic analysis. It examines an email header to identify suspicious relay servers, mismatched Reply-To addresses, and suspicious sender domain names. These checks help detect phishing emails and email spoofing attempts, illustrating a basic technique used in digital forensics and cybersecurity.